

Linear algebra

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(borrowed from *Math Tools for Neuroscience*, Prof. Jonathon Pillow,
Princeton University)

Linear algebra

“Linear algebra has become as basic and as applicable as calculus, and fortunately it is easier.”

- Gilbert Strang, *Linear algebra and its applications*

Topics

- vectors
 - addition
 - scalar multiplication
- vector norm
- unit vectors
- dot product
- linear projection
- **orthogonality**

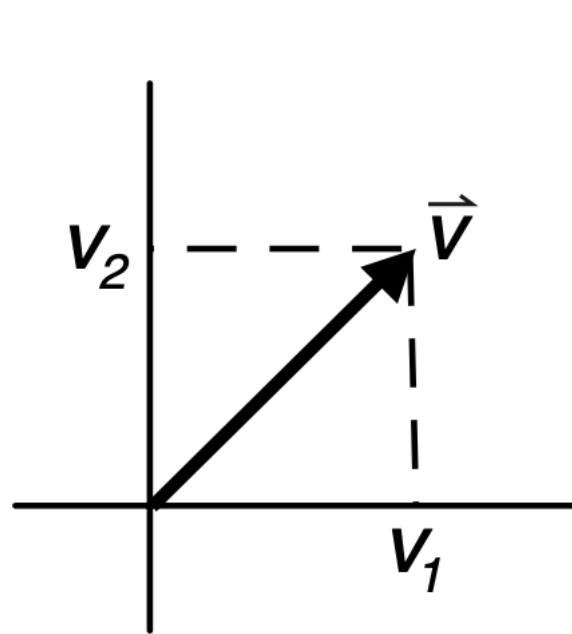
Vectors

a binary word [0,0,1,1,1,1,0,1, ...]

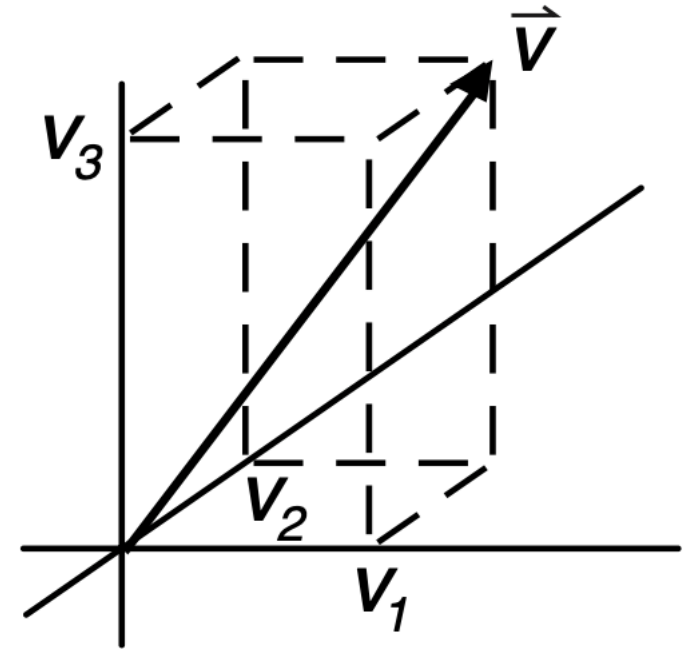
$$\xi = [1, 1, -1, 1, 1, -1, 1, 1, 1, 1]$$

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{pmatrix}$$

(N numbers)



(just 2 numbers)



(just 3 numbers)

column vector

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{pmatrix}$$

```
# make a 3-component column vector  
v = np.array([[3], [1], [-7]])
```

transpose

$$\vec{v}^T = (v_1 \ v_2 \ \cdots \ v_N)$$

```
# transpose  
v.T
```

row vector

Python

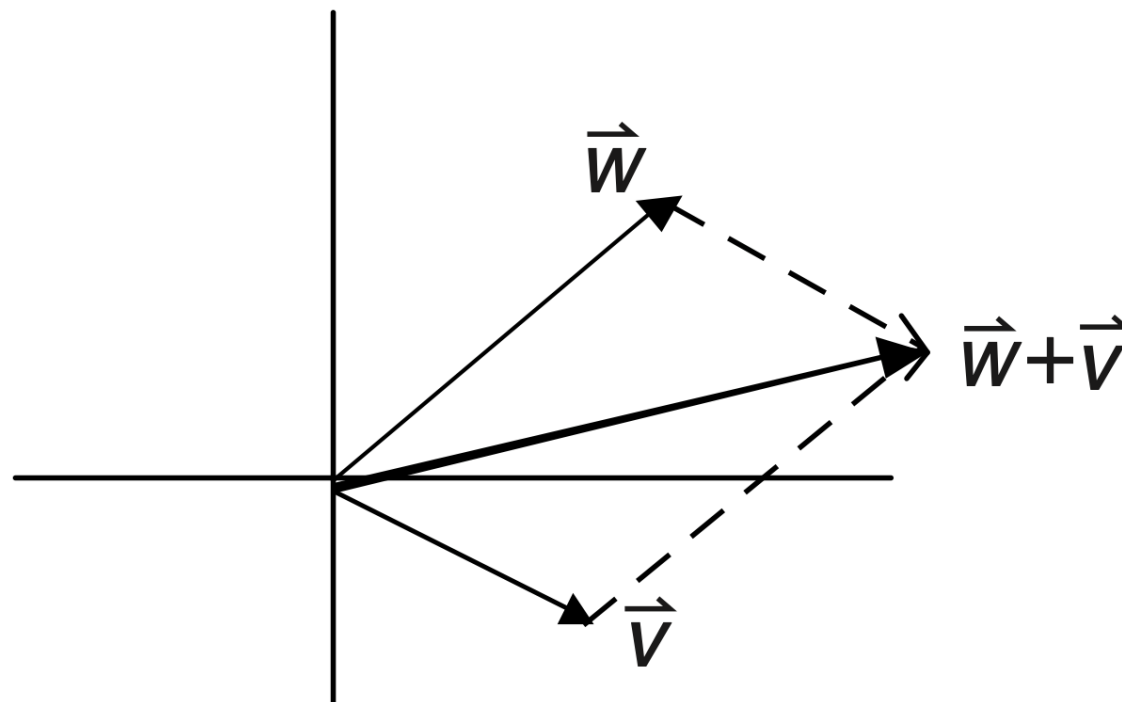
1. Vectors

```
# create row vector directly  
v = np.array([[3,1,-7]]) # row vector  
# or  
v = np.array([3,1,-7]) # 1D vector
```

Addition of vectors

$$\vec{v} + \vec{w} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} v_1 + w_1 \\ v_2 + w_2 \end{pmatrix}$$

Consider N = 2

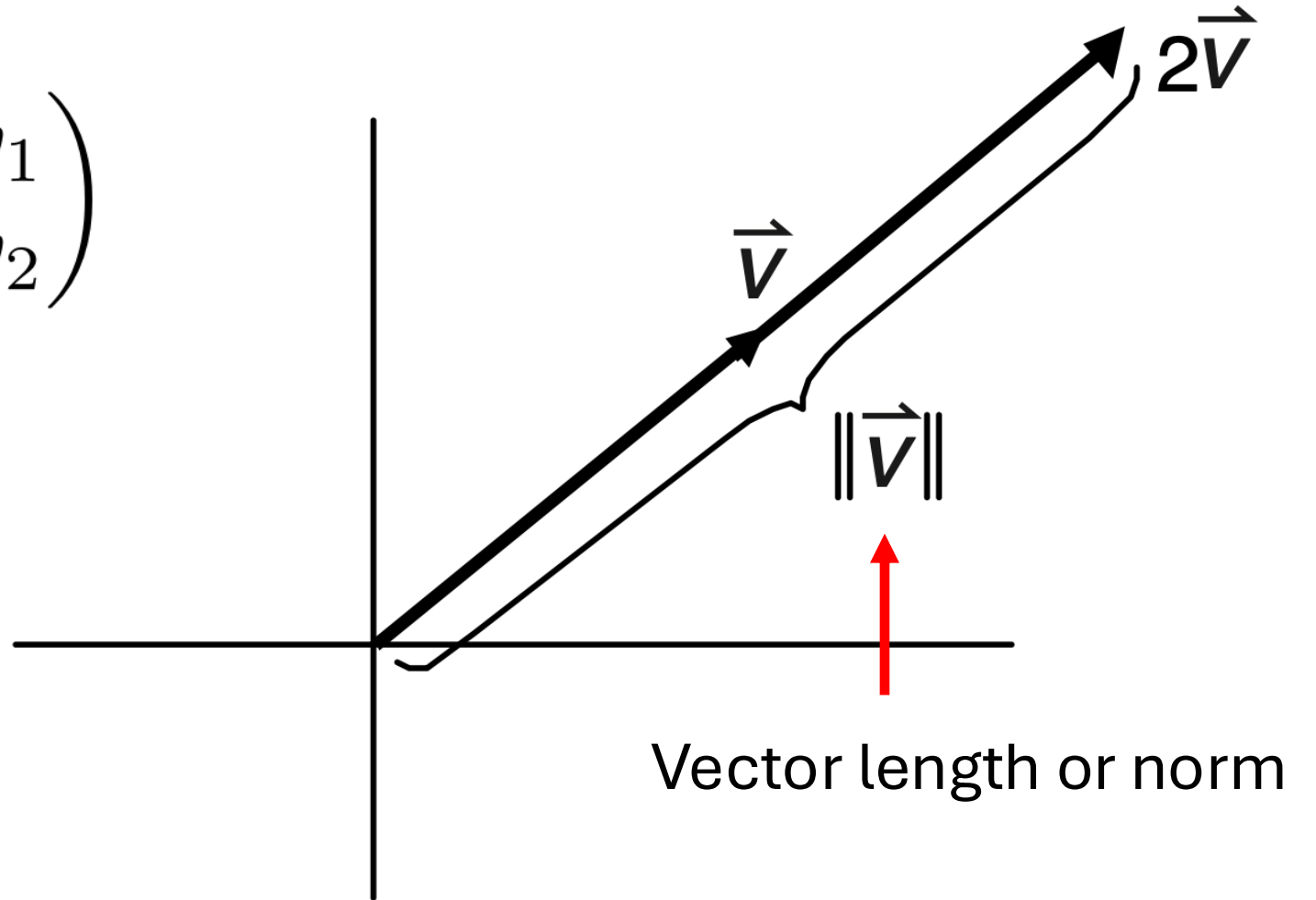


Scalar multiplication

Example $a = 2$

$$a\vec{v} = a \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} av_1 \\ av_2 \end{pmatrix}$$

Consider $N = 2$



Vector norm (“L2 norm”)

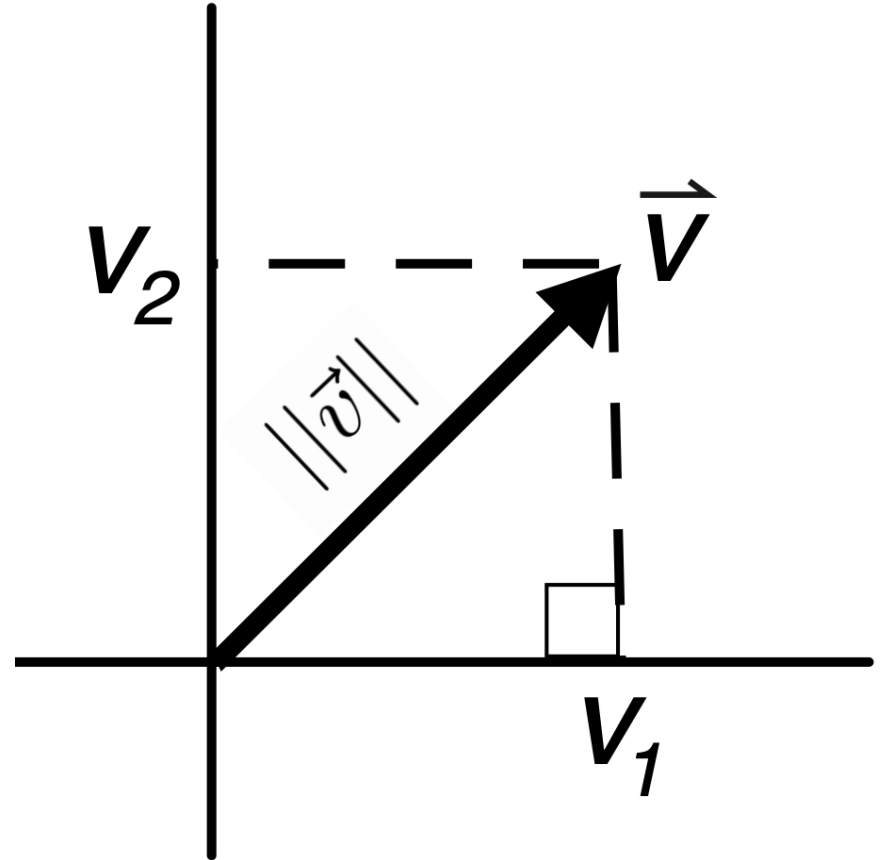
- vector length in Euclidean space

In 2-D:

$$||\vec{v}|| = \sqrt{v_1^2 + v_2^2}$$

In n -D:

$$||\vec{v}|| = \sqrt{v_1^2 + \dots + v_n^2}$$



Vector norm (“L2 norm”)

Exercises compute the vector length (norm of each of the following vectors):

- a) $[7, 7]$
- b) $[5, 5, 5, 5]$
- c) $[10, 1]$
- d) $[5, 5, 7]$

Vector norm (“L2 norm”) in Python

```
# make a vector
v = np.array([1, 7, 3, 0, 1])
```

```
# many equivalent ways to compute norm
np.linalg.norm(v)           # built-in function
np.sqrt(np.dot(v,v))        # sqrt of dot product
np.sqrt(v.T @ v)            # sqrt of v-transpose times v
np.sqrt(sum(v * v))          # sqrt of sum of elementwise product
np.sqrt(sum(v ** 2))         # sqrt of v elementwise-squared
```

```
# note use of @ and * and **
# @ - gives matrix multiply
# * - gives elementwise multiply
# ** - gives exponentiation (“raising to a power”)
```

Python

2. Vector Norm

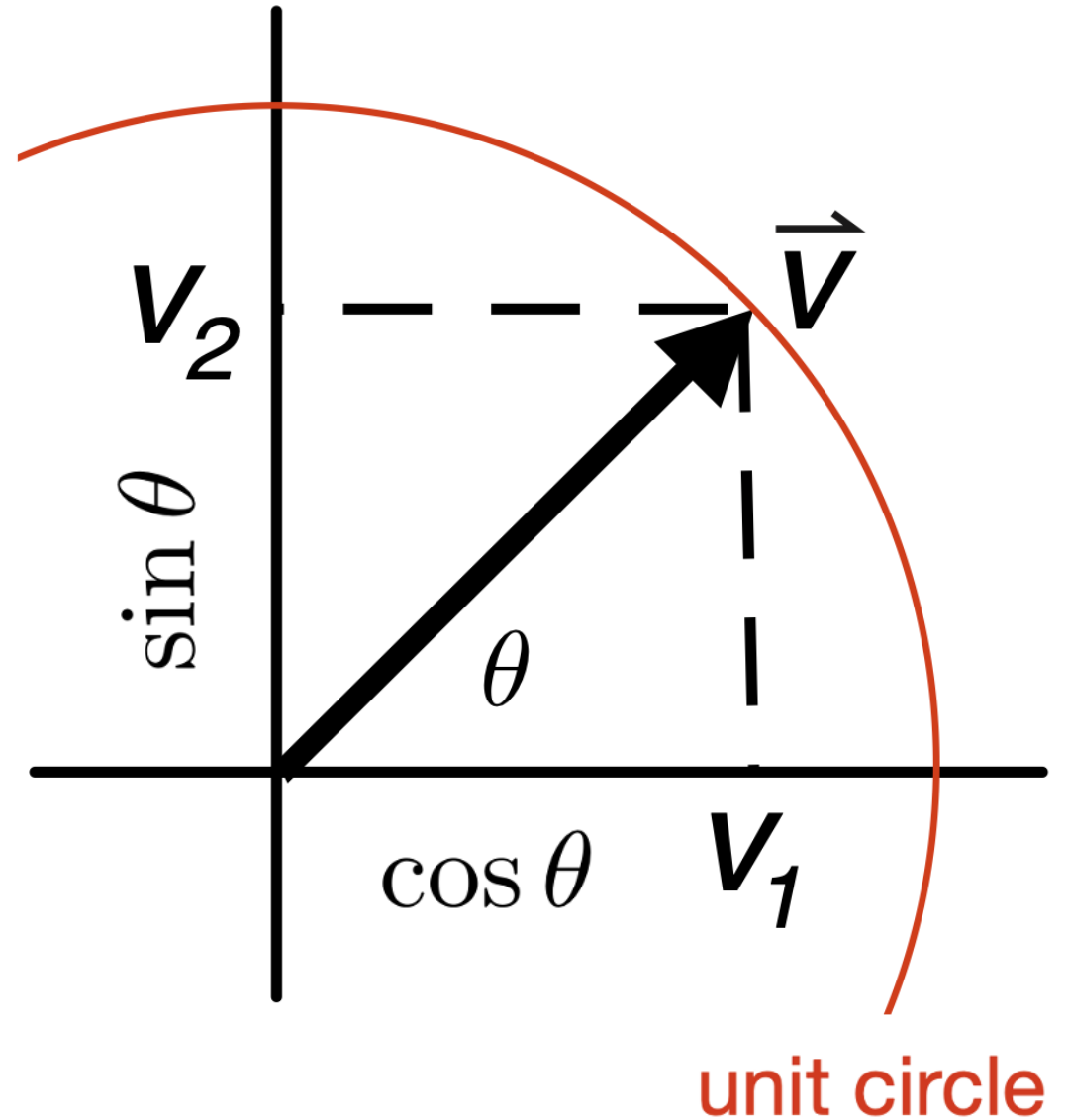
Unit vector

vector such that $||\vec{v}|| = 1$

in 2 dimensions

$$\vec{v} = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

$$\cos^2 \theta + \sin^2 \theta = 1$$



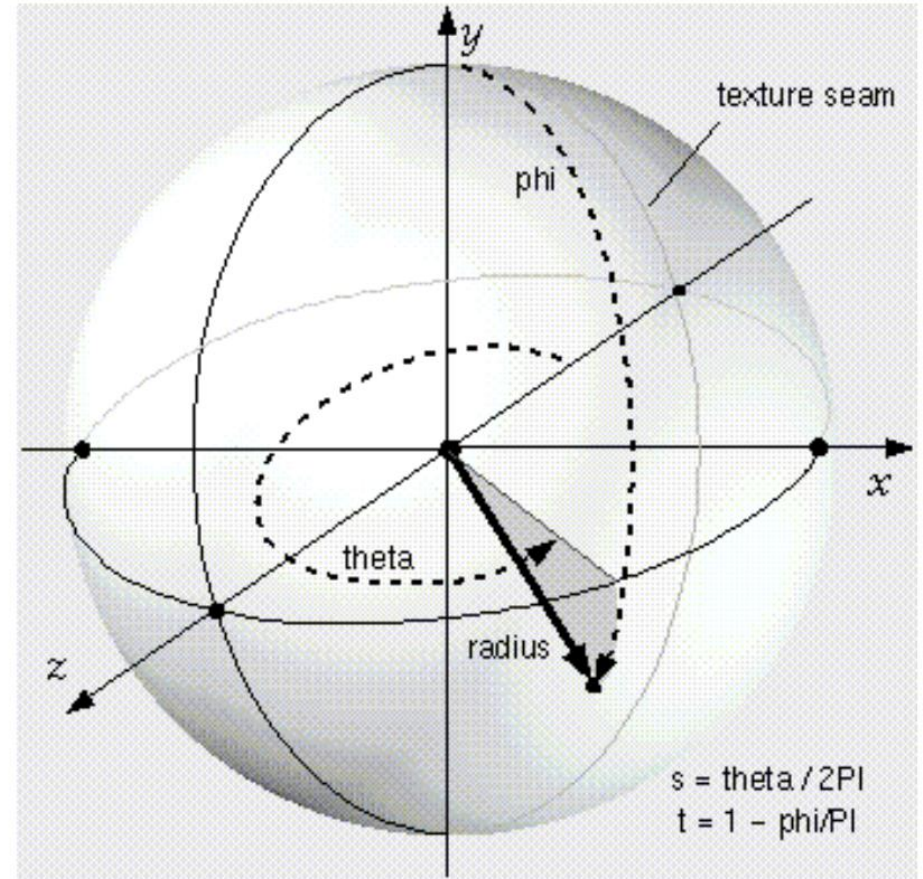
Unit vector

vector such that $||\vec{v}|| = 1$

in n dimensions

$$v_1^2 + v_2^2 + \dots + v_n^2 = 1$$

sits on the surface of an n -dimensional hypersphere



Unit vector

vector such that $||\vec{v}|| = 1$

We can make any vector into a unit vector

$$\vec{v} \rightarrow \frac{1}{||\vec{v}||} \vec{v}$$

Python

3. Unit Vector

Inner product (aka “dot product”)

- produces a scalar from two vectors

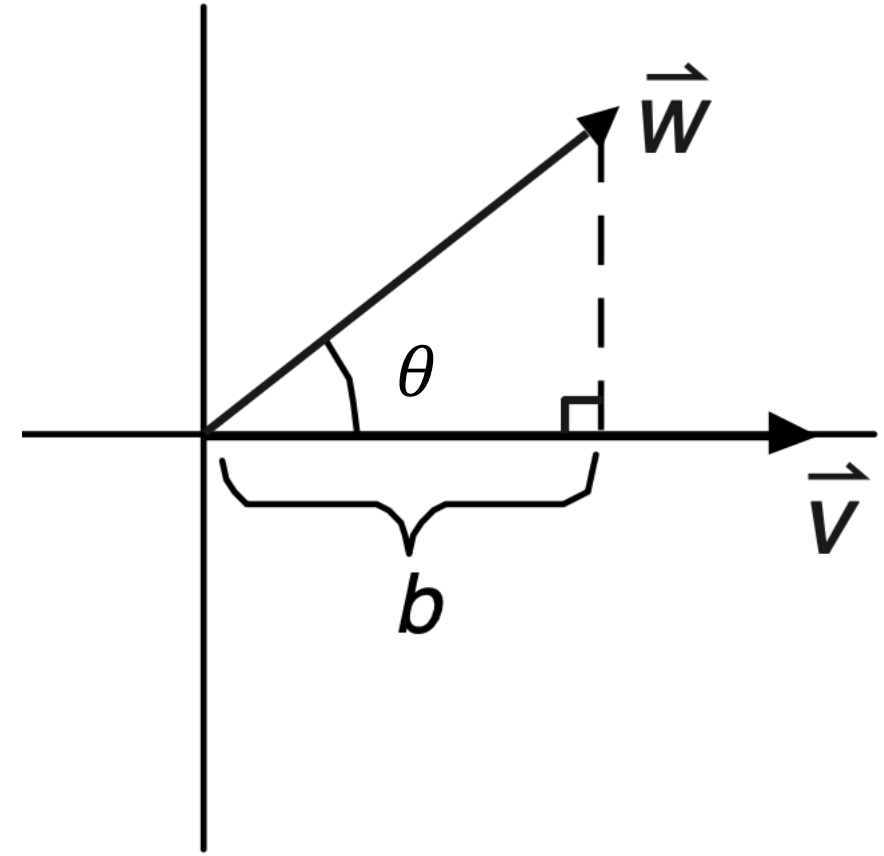
$$\vec{v} \cdot \vec{w}$$

$$\langle \vec{v}, \vec{w} \rangle$$

$$v_1 w_1 + v_2 w_2 + \cdots + v_n w_n$$

$$\|\vec{v}\| \|\vec{w}\| \cos \theta$$

$$\vec{v}^T \vec{w} = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$$



Inner product (aka “dot product”)

Exercises:

$v1 = [1, 2, 3]$

$v2 = [3, 2, -1]$

$v3 = [10, 0, 5]$

Compute:

$v1 \cdot v2$, $v1 \cdot v3$, $v2 \cdot v3$

Python

4. Inner product

Linear projection

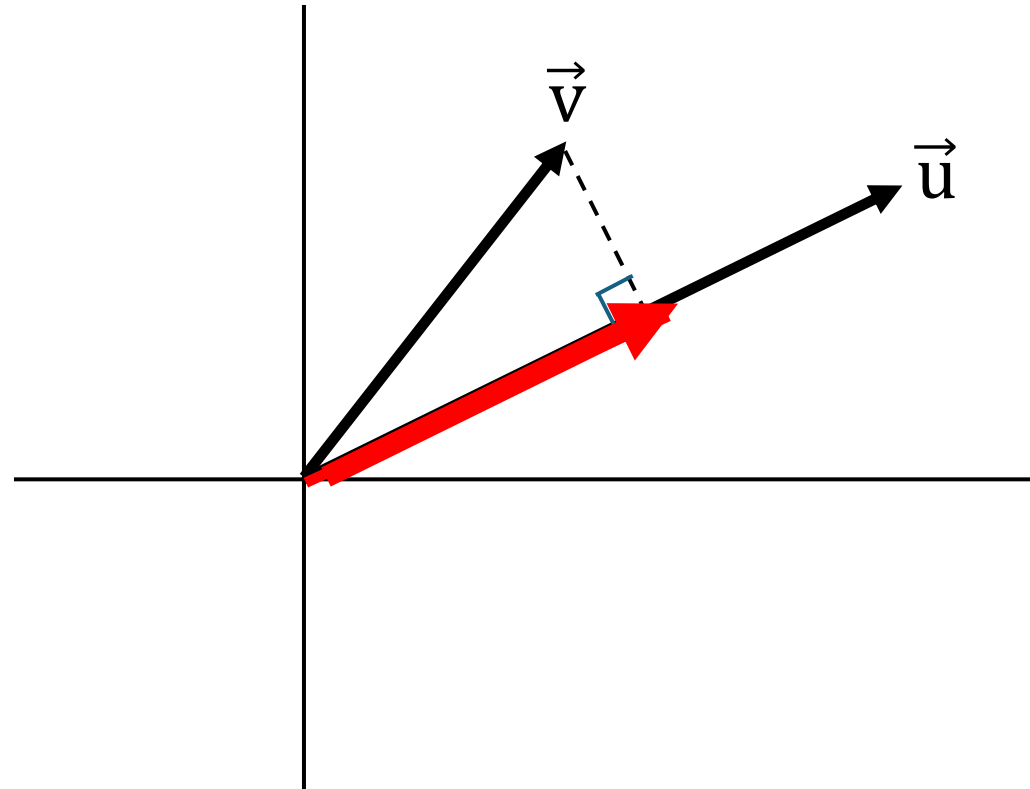
Intuitively, drop a vector down onto a line (or linear surface) at a right angle.

Projection of $\vec{v}$ onto $\vec{u}$

Length: $\frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}}$

Direction: $\vec{u}$

component of $\mathbf{v}$
in direction of $\mathbf{u}$ $\boxed{\frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}}$



Linear projection exercises

$$v = [2, 2]$$

$$u = [3, 1]$$

Compute projection of v onto u .

Python

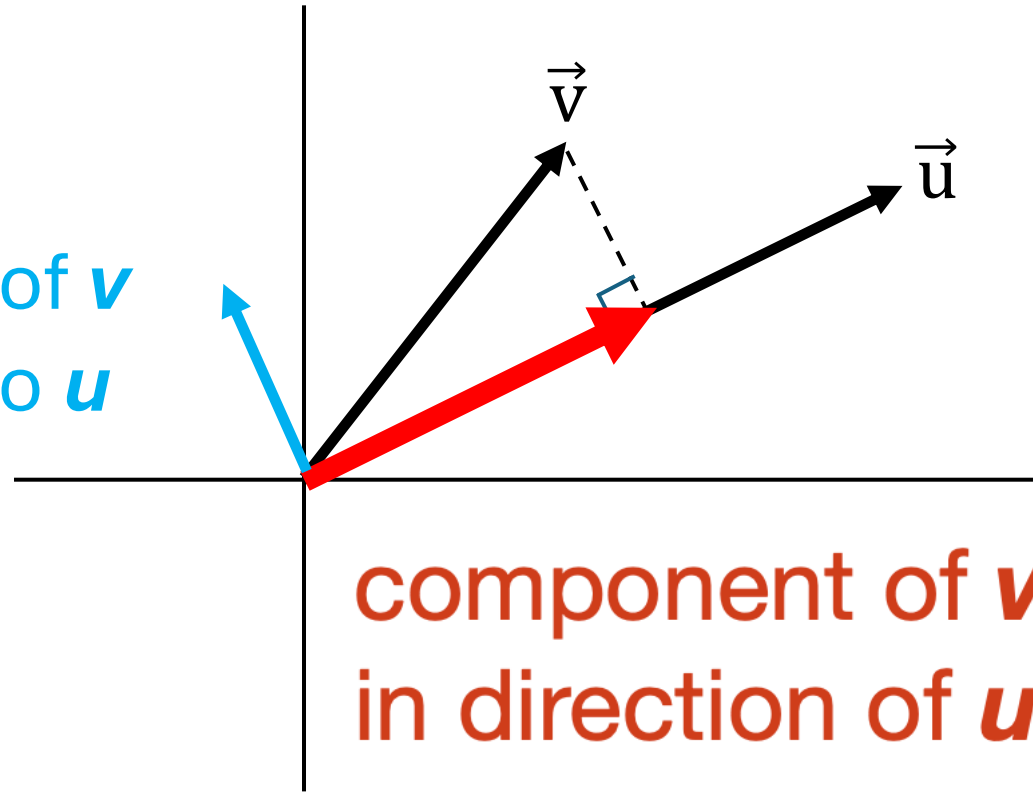
5. Linear projection

Orthogonality

Two vectors are orthogonal (or *perpendicular*) if their dot product is 0.

$$\vec{w} \cdot \vec{u} = 0$$

component of $\mathbf{v}$
orthogonal to $\mathbf{u}$



$$\vec{w} = \vec{v} - \frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}$$

component of $\mathbf{v}$
in direction of $\mathbf{u}$

$$\frac{\vec{v} \cdot \vec{u}}{\vec{u} \cdot \vec{u}} \vec{u}$$

Can decompose any vector $\mathbf{v}$ into its component along $\mathbf{u}$ and its orthogonal component.

Orthogonality exercises

Find a vector orthogonal to $x = [1, 0]$?

Let

$$v = [2, 2]$$

$$u = [3, 1]$$

Python

6. Orthogonality

Find the component of v orthogonal to u .

Topics

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- **orthogonality**